[INCH-POUND]
A-A-52554
27 January 1996
SUPERSEDING
MIL-H-370G
6 January 1994

COMMERCIAL ITEM DESCRIPTION

HOSE AND HOSE ASSEMBLIES, NONMETALIC: ELASTOMERIC, LIQUID FUEL

The General Services Administration has authorized the use of this commercial item description (CID) for all federal agencies.

1. SCOPE. This commercial item description (CID) applies to elastomeric, liquid fuel transfer hose and hose assemblies. The hose and hose assemblies are intended to be used for liquid fuel suction and discharge.

2. CLASSIFICATION. The hose and hose assemblies shall be of the following types, sizes, classes and fitting styles:

Type I	- Non-Wire Reinforced.
Type II	- Wire Reinforced, Noncollapsible.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data which may improve this document should be sent by letter to: US Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000.

FSC 4720

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 202610-26T09:13Z
Check the source to verify that this is the current version before use.

Size 1	-	0.50-in.
Size 2	-	0.75-in.
Size 3	-	1.00-in.
Size 4	-	1.25-in.
Size 5	-	1.50-in. ← (inside diameter)
Size 6	-	2.00-in.
Size 7	-	2.50-in.
Size 8	-	3.00-in.
Size 9	-	4.00-in.
Size 10	-	6.00-in.

Class 0	-	Bulk Hose (Without Fittings).
Class 1	-	Male Fitting On One End and Female Fitting On Other End.
Class 2	-	Male Fitting Both Ends.
Class 3	-	Female Fittings Both ends.

Style 0	-	Bulk Hose (Without Fittings).
Style A	-	Banded Shank (Reattachable).
Style B	-	Compression (Reattachable-Threaded).
Style D	-	Internal Expanded (One Time Threaded).

3. SALIENT CHARACTERISTICS

3.1 Description. The hose and hose assemblies shall consist of a tube, one or more plies of reinforcement, and a cover, for use with liquid hydrocarbon fuels.

3.2 Material. The materials used for the hose and hose assemblies are to be produced to comply with the provisions of this description, and meet with deterioration and corrosion resistance requirements. The use of recovered material made in compliance with regulatory requirements is acceptable providing that all requirements of this CID are met. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this description.

3.2.1 Tube. The basic material for the tube shall be a synthetic compound utilizing a copolymer product of butadiene and acrylonitrile or other synthetic fuel resistant elastomer that shall comply with all requirements of this CID.

3.2.2 Cover. The cover shall be elastomer compounded for fuel, oil, hydrolytic and ultraviolet stability and ozone resistance that complies with all requirements of this CID.

3.3 Reinforcement and static bond.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:13Z
Check the source to verify that this is the current version before use.

3.3.1 Type I. The reinforcement shall be one or more plies of braided, woven or spiraled synthetic fiber cord, evenly and firmly fabricated, and thoroughly impregnated with elastomeric compound, causing the plies to adhere firmly to each other and to the tube and cover and be of sufficient strength to assure conformance to the applicable requirements of this CID. The static bond shall be continuous to effect an electric bond between both ends of the hose (see 7.3). The bond shall not break when the hose is coiled or subjected to internal or external pressure.

3.3.2 Type II. The reinforcement shall have a helix or helixes of round, steel wire between the fiber reinforcement; or hose reinforcement may consist of a carcass of synthetic filament warp yarns interwoven with a helix or helixes of round, steel wire and fillers of cotton or synthetic filament yarns, alternately spared. The static bond shall be continuous to effect an electric bond between both ends of the hose (see 7.3). The bond shall not break when the hose is coiled or subjected to internal or external pressure.

3.4 Design and construction. The tube shall be constructed of a synthetic compound utilizing a copolymer product of butadiene and acrylonitrile or other synthetic fuel resistant elastomer. The tube shall be of uniform thickness not less than 0.08-inches (in.) and have an inside wall that is smooth and free from pits. The cover shall be elastomer compounded for fuel, oil, hydrolytic and ultraviolet stability and ozone resistant. The cover shall be of uniform thickness not less than 0.08-in. The cover shall be tan 686, in accordance with FED-STD-595, color chip 33446. The hose shall be constructed of an elastomeric inner tube, of type I or type II construction, and a compounded rubber cover.

3.5 Fittings (couplings). The fittings (couplings) shall be cast or forged aluminum. The fittings shall be protected with flexible dust plug(s) or cap(s) as appropriate.

3.5.1 Style A, banded shank. Unless otherwise specified (see 7.1), Style A fittings shall be cam-locking, quick-disconnect fittings conforming to MS27021 and MS27025.

3.5.2 Clamps. Hose clamps shall be type H of SAE J1508, and shall be austenitic stainless steel. Hose clamps shall be non-reflective. Two clamps shall be used for each fitting, except for the 6 in. hose, which shall have three clamps for each fitting. The widths of clamps shall be minimum 0.50 in. for hose size up to and including 2.50 in. inside diameter (I.D.) and 0.75 in. for hose 3.00 in., I.D. and up.

3.5.3 Style B. Reattachable compression fittings shall be two-piece construction. No cement, sealers, or adhesives shall be used in their installation other than to seal the cut ends of the hose. The body of the coupling shall be cast or forged with a smooth bore and shall include central hexagon or octagon wrench surfaces and a tailpiece consisting of two concentric walls with an intermediate hose chamber.

Source: https://assist.dla.mil -- Downloaded: 202610-26T09:13Z
Check the source to verify that this is the current version before use.

3.5.4 Style D. The one-time-use internal expanded fittings shall have an internal expanding serrated tailpiece, suitably gripping and compressing the wall of the hose against an external ferrule. When expanded, these fittings shall have an inside diameter conforming to the normal inside diameter of the hose.

3.6 Hose assemblies. Hose assemblies shall be type I or type II hose with fittings securely attached to the hose ~~to~~ form a permanent electric bond. The hose assembly shall not leak.

3.7 Diameters and tolerances of hose.

Inside Diameter (inches)	Outside Diameter (inches)
0.50 ± 0.0625	1.064 ± 0.0625
0.75 ± 0.0625	1.375 ± 0.03125
1.00 ± 0.0625	1.625 ± 0.0625
1.25 ± 0.0625	1.875 ± 0.0625
1.50 ± 0.0625	2.125 ± 0.0625
2.00 ± 0.0625	2.656 ± 0.0625
2.50 ± 0.0625	3.187 ± 0.0625
3.00 ± 0.0625	3.75 ± 0.0625
4.00 ± 0.0625	4.75 ± 0.0625
6.00 ± 0.0625	6.875 + 0.125 - 0.0625

3.8 Tests certification. Each length of hose or each hose assembly shall be certified as passing the hydrostatic proof pressure test in table I and having a continuous electrical bond. Certification shall be provided to demonstrate that a representative sample of hose and hose assemblies comply with all remaining tests as specified in this CID.

3.9 Hydrostatic and burst pressure. When subjected to the appropriate proof pressure in table I, the hose or hose assemblies shall not leak nor show any imperfections in either the hose or fittings, and there shall be no indication of slippage or pullout of the fitting from the hose. When subjected to the appropriate burst pressure in table I, the hose or hose assembly shall have no failure under the minimum burst pressure shown in table I. The length of the hose shall not change more than ±7 percent. The hose and hose assembly shall conform to the test pressures in table I.

Source: https://assist.dla.mil -- Downloaded: 202610-26T09:13Z
Check the source to verify that this is the current version before use.

TABLE I. Test loads and pressures.

Size of hose (inside diameter) (inches)	Proof test pressure (psi)	Burst test pressure (psi)	Working pressure (NTE) (psi)	Crush resistance (pounds)	Bend radius (inches)
0.50	150	500	125	200	2.75
0.75	150	500	125	200	3.50
1.00	150	500	125	200	4.00
1.25	150	500	125	200	5.00
1.50	150	500	125	225	6.00
2.00	125	400	100	250	8.00
2.50	125	375	94	325	10.00
3.00	100	300	75	325	12.00
4.00	100	300	75	325	16.00
6.00	100	300	75	325	24.00

3.10 Tensile strength and ultimate elongation. The tensile strength and ultimate elongation of the cover and tube of the hose and hose assemblies shall be determined in accordance with ASTM D471, using reference fuel B. The immersion temperature shall be 73 ± 1°F, with an immersion time of 70 hours(h). The tensile strength before fuel immersion shall not be less than 1250 pounds per square inch (psi). After fuel immersion, the tensile strength shall not be less than 40 percent or less than 600 psi (which ever is greater) of the tensile strength obtained before immersion. The ultimate elongation of the tube and cover before and after fuel immersion shall not be less than 200 and 100 percent respectively.

3.10.1 Volume increase. The volume increase of the tube and the cover after immersion in test fluid shall not exceed 60 and 100 percent respectively.

3.11 Original adhesion. The original adhesion between the tube and plies, between the plies, and between the cover and plies before and after fuel immersion shall be 10 and 6 pounds (lbs) respectively.

3.12 Low temperature flexibility. Low temperature flexibility for the cover and tube shall be at a test temperature of 90 ± 2 °F. The exposure time shall be 7 days, and when tested, the cover and tube shall not exceed a Young's Modulus of 10,000 psi.

3.13 Existent gum. A hose not less than 14 inches long shall be stoppered at one end with a 2 in. plug and secured with a clamp. The hose shall be filled with fuel and similarly secured at the opposite end. The hose shall be tested for a period of seven days by agitating the fuel by means

Source: https://assist.dla.mil -- Downloaded: 202610-26T09:13Z
Check the source to verify that this is the current version before use.

of movement every 24 h period. The agitation process is performed by moving the hose from the vertical position to a horizontal position at a rate of 2 cycles per minute (min) for 5 min at each 24 h period. The fuel shall be tested before and after in accordance with ASTM D381. The difference in the two fuels is referred to as the existent gum residue extracted with haptene, and this shall not be more than 0.002 lbs per gallon (gal).

3.14 Crush resistance (type II only). Crush resistance shall be determined by centering a 12 in. length of hose between 3 in. wide, parallel metal plates in such a way that a 3 in. length of hose is being compressed. The plates shall be brought together at a rate of 2 in. per min until the applicable crush resistance load specified in table I has been applied. The distance between plates shall be measured with a steel rule and expressed in percent of the original outside diameter. The load shall be released and the minimum outside diameter of the hose at the center of the compressed area shall be measured and reported in percent of original outside diameter. The smallest outside diameter shall not be less than 85 percent of the original outside diameter. After release of the load, the smallest outside diameter shall not be less than 95 percent of the original outside diameter.

3.15 Ozone resistance. After conditioning for 24 h in an ozone-free atmosphere, the cover shall be exposed for 72 h at 104 $\pm$ 1°F, to an atmosphere containing 50 parts, $\pm$ 10 parts, per hundred million (pphm) of ozone. At the end of the exposure time, there shall be no visible cracking in the cover when viewed under 7X magnification.

3.16 Bend radius. A test specimen shall be subjected to the bend radius and proof test pressure in table I. After this process is completed, there shall be no pressure loss.

4. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. QUALITY ASSURANCE PROVISIONS

5.1 Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this CID and that the product conforms to the producers drawing, specifications, workmanship standards and quality assurance practices. Items with known defects shall not be submitted for Government acceptance. The government reserves the right to require proof of such conformance prior to acceptance. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:13Z
Check the source to verify that this is the current version before use.

6. PACKAGING

Preservation, packing, and marking for the desired level shall be as specified in the contract or order (see 7.1).

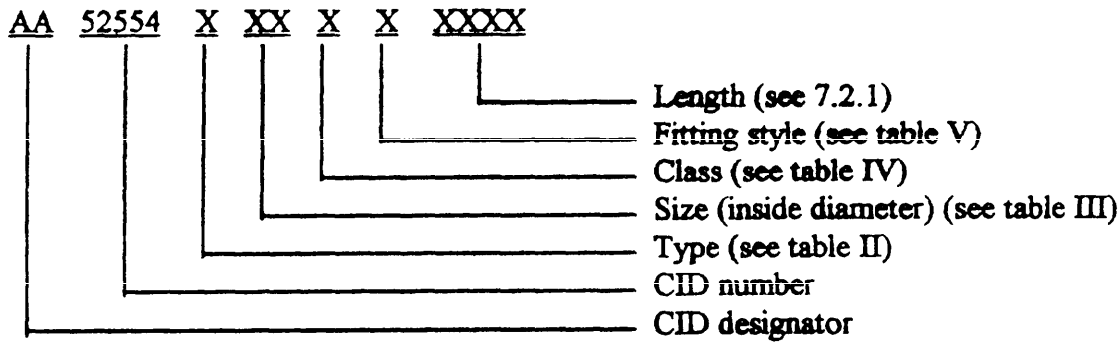
7. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

7.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this CID.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced.
c. Style A fitting if other than specified (see 3.5.1).
d. Part identification number (PIN) and quantity required.
e. Selection of applicable level of packaging required.

7.2 Part or identification number (PIN). The part or identification number (PIN) of the hose and hose assemblies covered by this CID, shall be designed in the following form (see 7.1).



Example: AA52554-B08B21700 Hose assembly, in accordance with MIL-H-370, type II (type II - wire reinforced), size 8 (3 in.), class 1 (male fitting one end, female fitting on the other end), style A (banded shank reattachable), length (170.0 in. long).

7.2.1 Length. The length of hose or hose assembly shall be specified ± 2 percent, exclusive of fitting and written in clear text and will be in inches with the last number expressed in tenths of an inch.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:13Z
Check the source to verify that this is the current version before use.

7.2.2 Classification. The hose and hose assemblies shall be the types (see table II), sizes (inside diameter) (see table III), classes (see table IV), styles (see table V), and lengths (see 7.2.1), as specified.

TABLE II. Type designator.

Symbol	Characteristics
A	Type I - Non-wire reinforced
B	Type II - Wire reinforced, noncollapsible

7.2.3 Size. The hose and hose assembly size is identified by a two digit code number as shown in table III.

TABLE III. Size designator.

Code No.	Inside Diameter (inches)
01	Size 1 - 0.500
02	Size 2 - 0.750
03	Size 3 - 1.00
04	Size 4 - 1.25
05	Size 5 - 1.50
06	Size 6 - 2.00
07	Size 7 - 2.50
08	Size 8 - 3.00
09	Size 9 - 4.00
10	Size 10 - 6.00

7.2.4 Class. The hose assembly class is identified by a one letter symbol as shown in table IV.

TABLE IV. Class designator.

Symbol	Characteristics
A	Class 0 - Bulk Hose (without fittings).
B	Class 1 - Male fitting on one end and female fitting on other end.
C	Class 2 - Male fitting both ends.
D	Class 3 - Female fittings both ends.

7.2.5 Fitting. The hose assembly fitting style is identified as a single code numeral as shown in table V.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:13Z
Check the source to verify that this is the current version before use.

TABLE V. Fitting designator.

Code No.	Characteristics
1	Style 0 - Bulk Hose (without fittings).
2	Style A - Banded shank (reattachable).
3	Style B - Compression (reattachable-threaded).
4	Style D - Internal expanded (one time threaded).

7.3 Electrical bond. An electrical bond eliminates a difference in potential between objects.

7.4 Regulatory requirements. Recovered material is material made in conformance to section 402 of the Clean Air Act (33 U.S.C. 1343 et. Seq.).

7.5 Government documents. Copies of FED-STD-595 "Colors Used in Government Procurement", MS27021 and MS27025, "Coupling Half, Quick Disconnect, Cam-Locking Type Male/Female, Hose Shank Type II/Type VI" are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.

7.6 Non-Government documents. Copies of ASTM D381, "Standard Test Methods for Existent Gum in Fuels by Jet Evaporation" and ASTM D471, "Standard Test Methods for Rubber Properties-Effects of Liquids" are available from the American Society for Testing and Materials (ASTM), 1916 Race Street, Philadelphia, PA 19103-1187. Copies of SAE J1508 "Hose Clamp Specification, Standards" are available from the Society of Automotive Engineers (SAE), Inc., 400 Commonwealth Drive, Warrendale, PA 15096-0001.

MILITARY INTERESTS:

Custodians:
Army - AT
Navy - YD1
Air Force - 99

Review Activities:
Navy - SH, MC
Air Force - 82
DLA - CS

CIVIL AGENCY COORDINATING ACTIVITY:
GSA - FSS

PREPARING ACTIVITY:
Army - AT

(Project 4720-0085)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:13Z
Check the source to verify that this is the current version before use.